

1

Galaxy Europe

Using 11.3 MB Login Register

Upload Download from URL or upload files from disk

Tools

Get Data

Send Data

Collection Operations

GENERAL TEXT TOOLS

Text Manipulation

Convert Formats

Filter and Sort

Join, Subtract and Group

GENOMIC FILE MANIPULATION

Convert Formats

FASTA/FASTQ

FASTQ Quality Control

Quality Control

SAM/BAM

BED

VCF/BCF

Nanopore

COMMON GENOMICS TOOLS

Operate on Genomic Intervals

Fetch Sequences / Alignments

Manage windows in Galaxy

"Anyone, anywhere in the world should have free, unhindered access to not just my research, but to the research of every great and enquiring mind across the spectrum of human understanding." – Prof. Stephen Hawking

News More news

Galactic Weekly: Week 12, 2026

Weekly summary of activity across 150+ galaxyproject repositories

Events More events

Small Scale Galaxy Admins Meeting

Apr 16 · Online, Global

European Geosciences Union (EGU) 2026

May 4 · Vienna

A Data Terra session to the European Geoscience Union

History

search datasets

Unnamed history

Default Storage

11.8 MB

This history is empty. You can load your own data or get data from an external source.

Galaxy Europe

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GENOMIC FILE MANIPULATION

Convert Formats

FASTA/FASTQ

FASTQ Quality Control

Quality Control

Upload from Disk or Web to Unnamed history

Regular Composite Collection Rule-based

final_data.vcf 11.3 MB Auto-detect unspecified (?) 100%

Type (set all): Auto-detect Reference (set all): unspecified (?)

Choose local file Choose from repository Paste/Fetch data Start Pause Reset Close

History

search datasets

Unnamed history

Default Storage

11.8 MB

2: final_data.vcf

2

Galaxy Europe

Using 11.3 MB Login Register

Tools

Discover Tools

stacks2 populations

Show Sections

Stacks2: populations Calculate population-level summary statistics

Stacks2: populations Calculate population-level summary statistics (Galaxy Version 2.55+galaxy4)

Run Tool

Tool Parameters

Input type

VCF file

select input file type

VCF file *

1: final_data.vcf

accepted formats

(-V)

Specify a population map - optional

Nothing selected

accepted formats

Format is 'SAMPLE1 \t POP1 \n SAMPLE2 ...' (-M)

History

search datasets

Unnamed history

Default Storage

11.8 MB

1: final_data.vcf

Galaxy Europe

Using 11.3 MB Login Register

Tools

Discover Tools

stacks2 populations

Show Sections

Stacks2: populations Calculate population-level summary statistics

Stacks2: populations Calculate population-level summary statistics (Galaxy Version 2.55+galaxy4)

Run Tool

Output options

If data is reference aligned, exports will be ordered; only a single representative of each overlapping site.

No

(--ordered-export)

Output locus consensus sequences in FASTA format

No

(--fasta-loci)

Output the sequences of the two haplotypes of each (diploid) sample, for each locus, in FASTA format

No

(--fasta-samples)

Output results in Variant Call Format (VCF)

No

(--vcf)

Output results in GenePop Format

No

(--genepop)

Output results in Structure Format

Yes

(--structure)

History

search datasets

Unnamed history

Default Storage

11.8 MB

1: final_data.vcf

Galaxy

7

Discover Tools

remove

Show Sections

Remove beginning of a file

Remove sequencing artifacts

Remove Description from fasta file

Remove coordination complexes Remove molecular coordination complexes from a list of structure representations

Remove small molecules from a library of compounds

Remove coordinates and fragments from

Remove beginning of a file (Galaxy Version 1.0.0)

Run Tool

Tool Parameters

Remove first *

1

lines

from *

8: Stacks2: populations on dataset 1 Structure format

accepted formats

Additional Options

Email notification

No

Send an email notification when the job completes.

8

search datasets

Unnamed history

Default Storage

38.7 MB

8

8: Stacks2: populations on data set 1 Structure format

7: Stacks2: populations on data set 1 Population-level summary statistics

Galaxy

9

Discover Tools

structure

Show Sections

Structure using multi-locus genotype data to investigate population structure

structureHarvester for parsing STRUCTURE outputs and for performing the Evanno method

Structure to GSPAN

ChEMBL structure pipeline for curation and standardizing of molecular structures

GROMACS structure configuration using editconf

Align structures and compute relative

Structure using multi-locus genotype data to investigate population structure (Galaxy Version 2.3.4+galaxy1)

Run Tool

Tool Parameters

Genotype data *

9: Remove beginning on dataset 8

accepted formats

Number of runs *

1

Note that the runs are sequential. Please launch separate runs if it's too long

mainparams

! Please provide a value for this option.

Number of populations assumed * required

or [K] (MAXPOPS)

History

search datasets

Unnamed history

Default Storage

50.1 MB

9

9: Remove beginning on dataset 8

8: Stacks2: populations on data set 1 Structure format

7: Stacks2: populations on data set 1 Population-level summary statistics

STRUCTURE parameters

1. Number of populations assumed = 2
2. Length of burnin period = 1000
3. Number of MCMC reps after burnin = 2000
4. Number of diploid individuals in data file = 170
5. Number of loci in data file = 16548
6. Input file contains a user-defined population-of-origin for each individual = No
7. Number of additional columns of data before the genotype data start. = 1
8. In “extraparams” => Use POPDATA for location information = No

10

13: run_K_2.extraparams



12: run_K_2.mainparams



11: run_K_2.log

a list with 1 **txt** dataset

10: run_K_2.out

a list with 1 **txt** dataset

11

1: K_2_run_1_f



98,152 lines

format **txt**, database ?

STRUCTURE by Pritchard, Stephens and Donnell
and Falush, Stephens and Pritchard (2003)

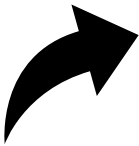
54 Inferred ancestry of individuals:

	Label (%Miss)	Inferred clusters
55		
56	1 FF68_CN1 (0)	: 0.119 0.881
57	2 FF68_CN10 (0)	: 0.000 1.000
58	3 FF68_CN100 (0)	: 0.000 1.000
59	4 FF68_CN101 (0)	: 0.000 1.000
60	5 FF68_CN102 (0)	: 0.000 1.000
61	6 FF68_CN103 (0)	: 0.000 1.000
62	7 FF68_CN104 (0)	: 0.000 1.000
63	8 FF68_CN105 (0)	: 0.001 0.999
64	9 FF68_CN106 (0)	: 0.000 1.000
65	10 FF68_CN107 (0)	: 0.000 1.000

...

215	160	FF68_CN9	(0)	:	0.609 0.391
216	161	FF68_CN90	(0)	:	0.413 0.587
217	162	FF68_CN91	(0)	:	0.520 0.480
218	163	FF68_CN92	(0)	:	0.150 0.850
219	164	FF68_CN93	(0)	:	0.799 0.201
220	165	FF68_CN94	(0)	:	0.000 1.000
221	166	FF68_CN95	(0)	:	0.000 1.000
222	167	FF68_CN96	(0)	:	0.000 1.000
223	168	FF68_CN97	(0)	:	0.000 1.000
224	169	FF68_CN98	(0)	:	0.000 1.000
225	170	FF68_CN99	(0)	:	0.000 1.000

Copy to excel



Inferred	ancestry	of	individuals:			
	Label	(%Miss)	:	Inferred	clusters	
	1	FF68_CN1	0 :		0.119	0.881
	2	FF68_CN10	0 :		0	1
	3	FF68_CN100	0 :		0	1
	4	FF68_CN101	0 :		0	1
	5	FF68_CN102	0 :		0	1
	6	FF68_CN103	0 :		0	1
	7	FF68_CN104	0 :		0	1
	8	FF68_CN105	0 :		0.001	0.999
	9	FF68_CN106	0 :		0	1
	10	FF68_CN107	0 :		0	1
	11	FF68_CN108	0 :		0	1
	12	FF68_CN109	0 :		0	1
	13	FF68_CN11	0 :		0	1
	14	FF68_CN110	0 :		0	1

Rearrange the file:

1. Remove the first two rows
2. Cut B1:G99
3. Paste at A1
4. Copy Column B, paste in C
5. Remove column A and D
6. Insert first row
7. Type column names
8. save to ".csv" file



54 ▾	Inferred ancestry of individuals:					
55		Label	(%Miss)	:	Inferred	clusters
56	1	FF68_CN1	(0)	:	0.119	0.881
57	2	FF68_CN10	(0)	:	0.000	1.000
58	3	FF68_CN100	(0)	:	0.000	1.000
59	4	FF68_CN101	(0)	:	0.000	1.000
60	5	FF68_CN102	(0)	:	0.000	1.000
61	6	FF68_CN103	(0)	:	0.000	1.000
62	7	FF68_CN104	(0)	:	0.000	1.000
63	8	FF68_CN105	(0)	:	0.001	0.999
64	9	FF68_CN106	(0)	:	0.000	1.000
65	10	FF68_CN107	(0)	:	0.000	1.000

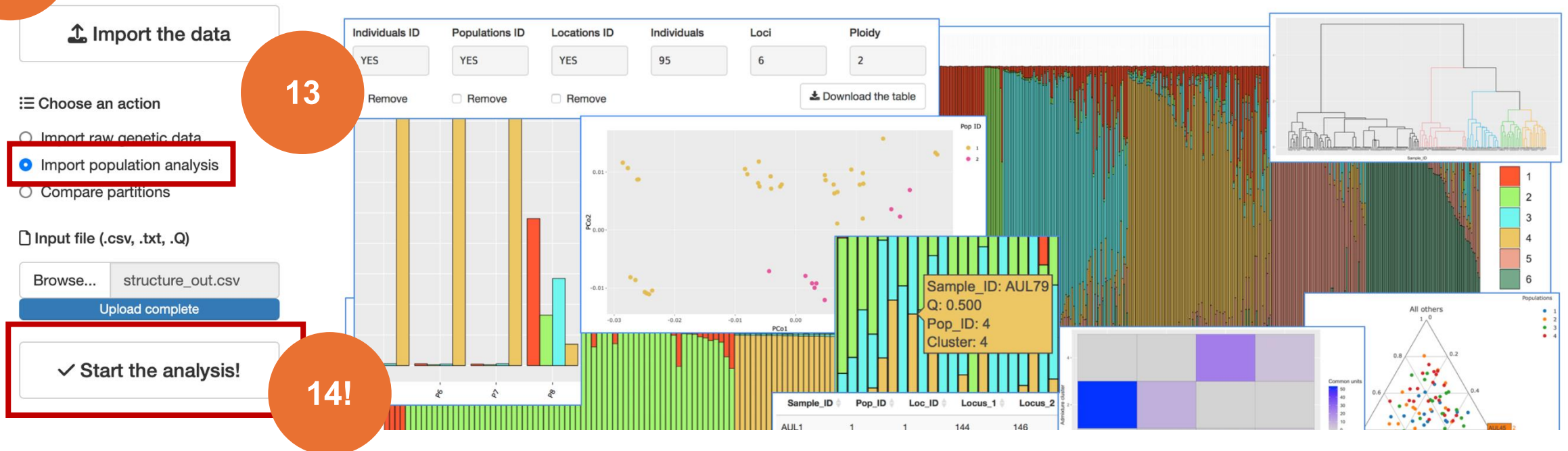
Sample_ID	Pop_ID	K1	K2
FF68_CN1	FF68_CN1	0.119	0.881
FF68_CN10	FF68_CN10	0	1
FF68_CN100	FF68_CN100	0	1
FF68_CN101	FF68_CN101	0	1
FF68_CN102	FF68_CN102	0	1
FF68_CN103	FF68_CN103	0	1
FF68_CN104	FF68_CN104	0	1
FF68_CN105	FF68_CN105	0.001	0.999
FF68_CN106	FF68_CN106	0	1
FF68_CN107	FF68_CN107	0	1

StruRly

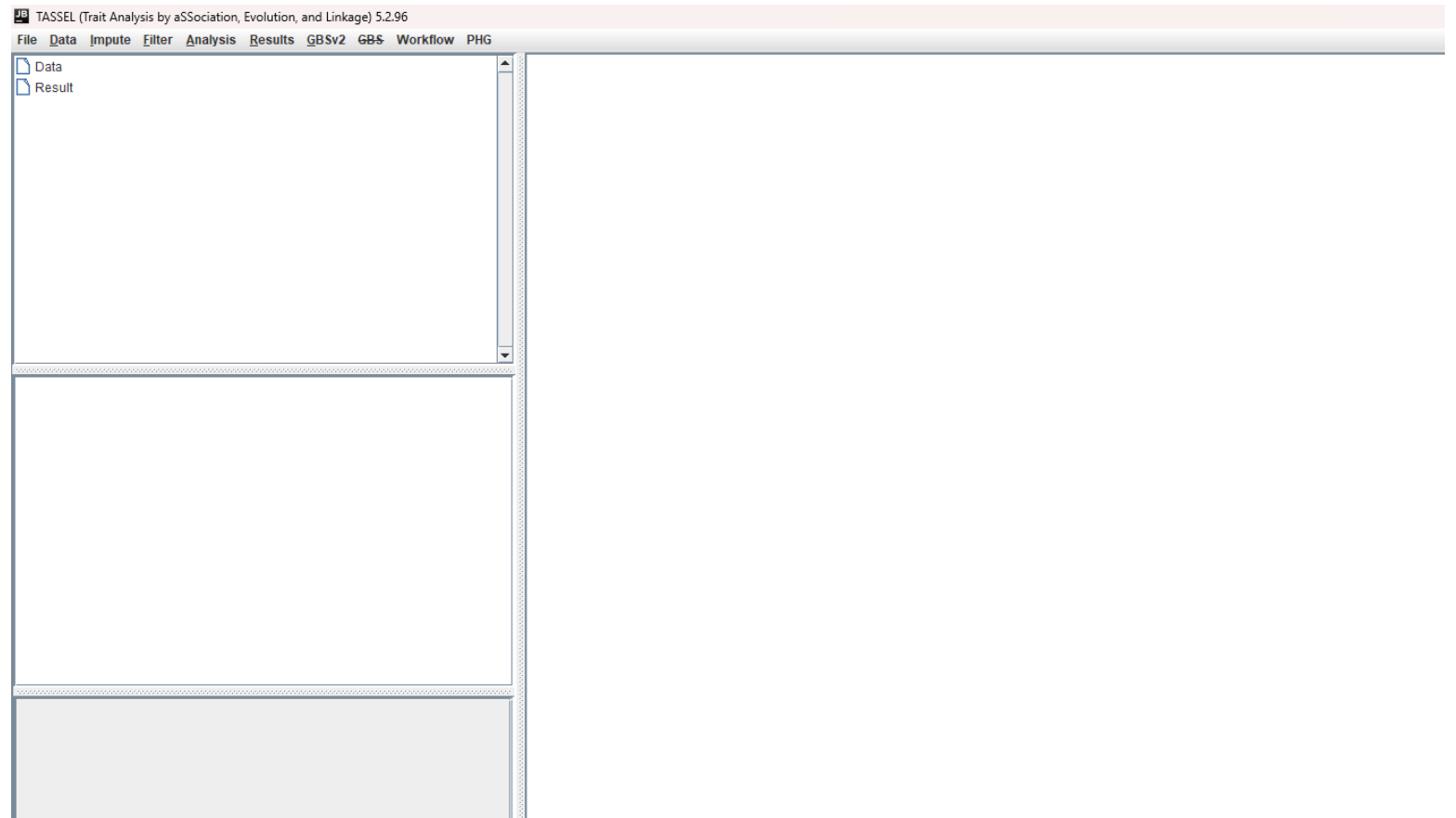
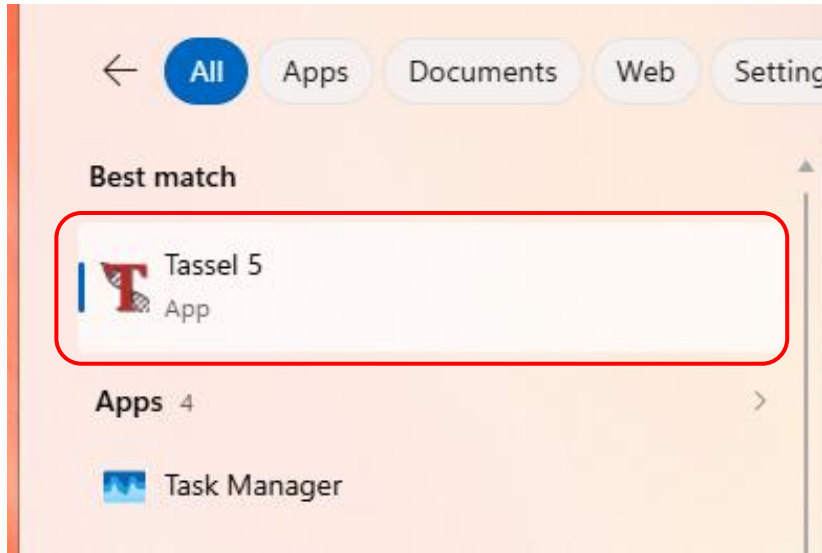
<https://nicocriscuolo1618.shinyapps.io/StruRly/>

12

StruRly 0.1.0 - Comprehensive, detailed and interactive plots for *STRUCTURE* and *ADMIXTURE* population analysis

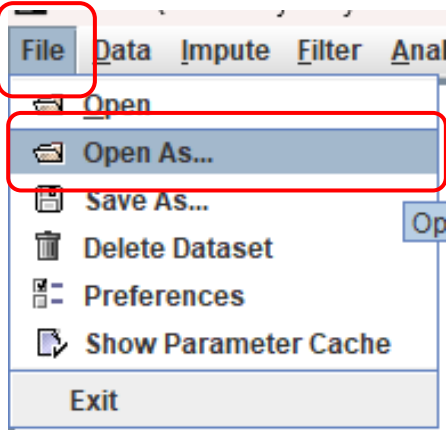


Open Tassel

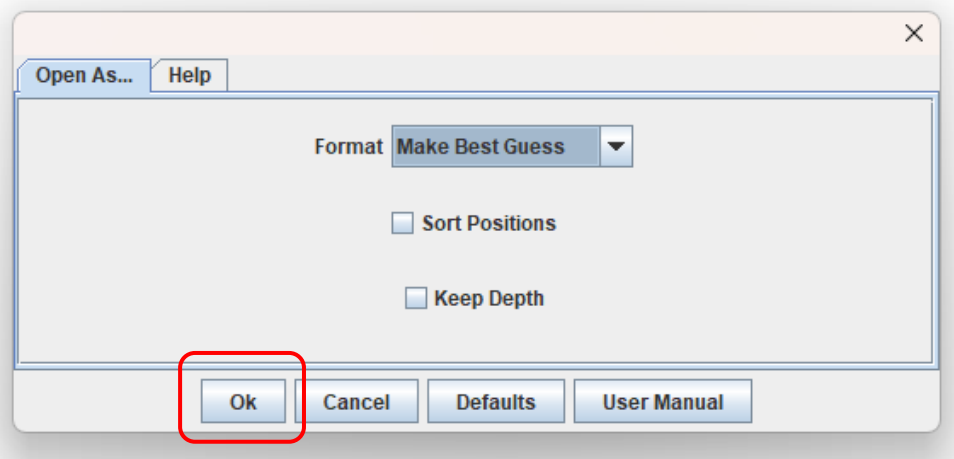


Input Genotype file (.vcf)

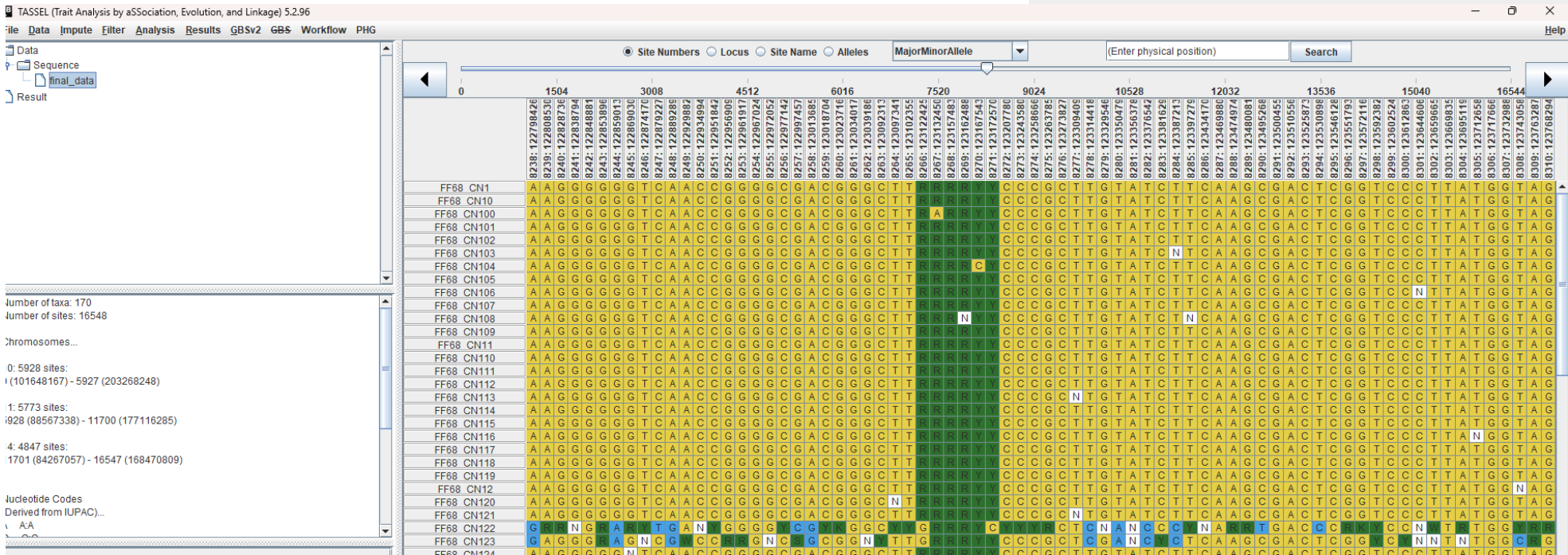
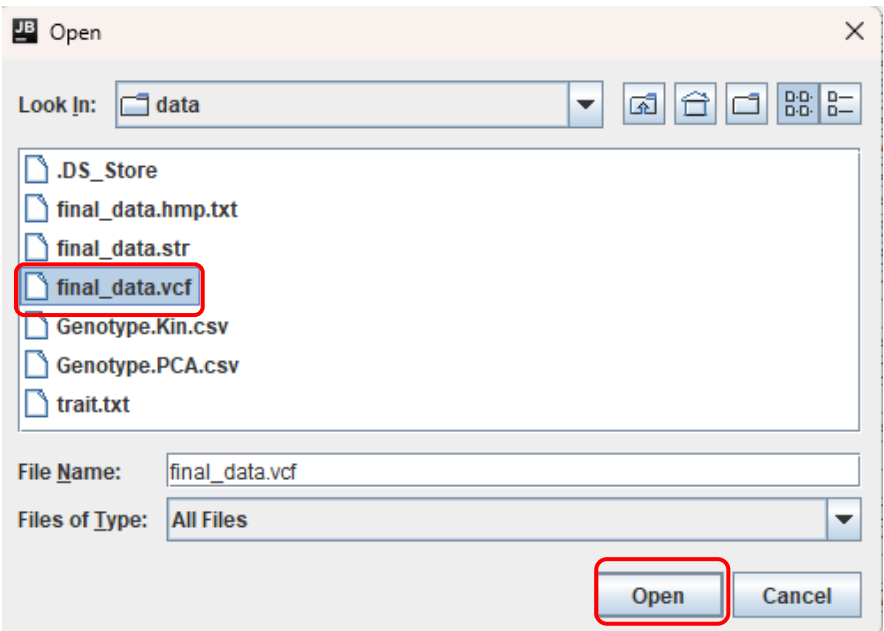
2.1



2.2

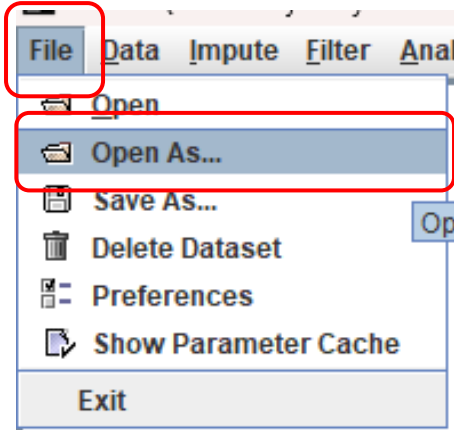


2.3

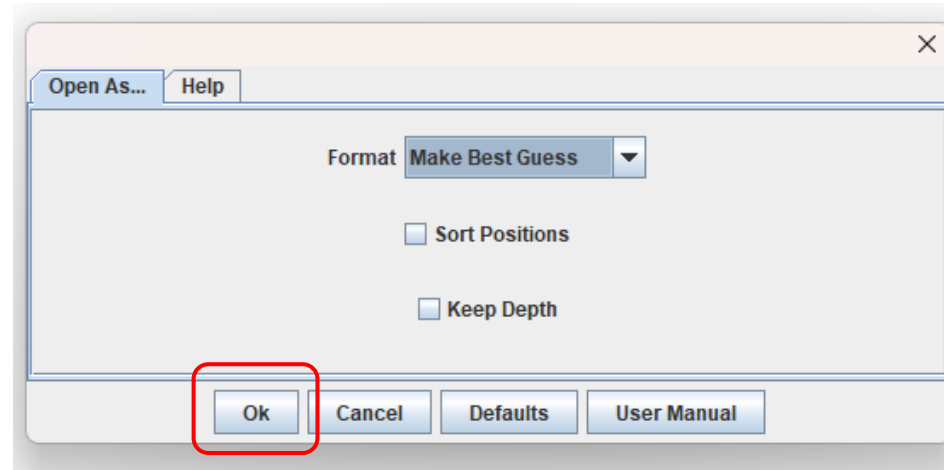


Input phenotype file (.txt)

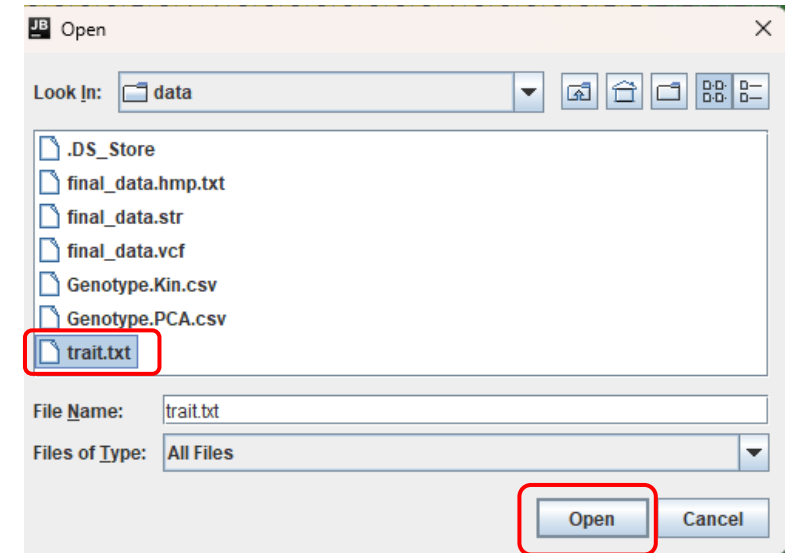
3.1



3.2



3.3



TASSEL (Trait Analysis by aSSociation, Evolution, and Linkage) 5.2.96

File Data Impute Filter Analysis Results GBSv2 GBS Workflow PHG

Data

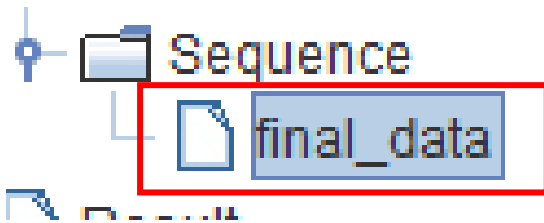
- Numerical
 - trait
- Sequence
 - final_data
- Result

Taxa	Color
FF68_CN1	1
FF68_CN2	1
FF68_CN3	3
FF68_CN4	5
FF68_CN5	5
FF68_CN6	1
FF68_CN7	1
FF68_CN8	1
FF68_CN9	1
FF68_CN10	1
FF68_CN11	1
FF68_CN12	1
FF68_CN13	2
FF68_CN14	5
FF68_CN15	1
FF68_CN16	1
FF68_CN17	1
FF68_CN18	1
FF68_CN19	1
FF68_CN20	1
FF68_CN21	1
FF68_CN22	1
FF68_CN23	2

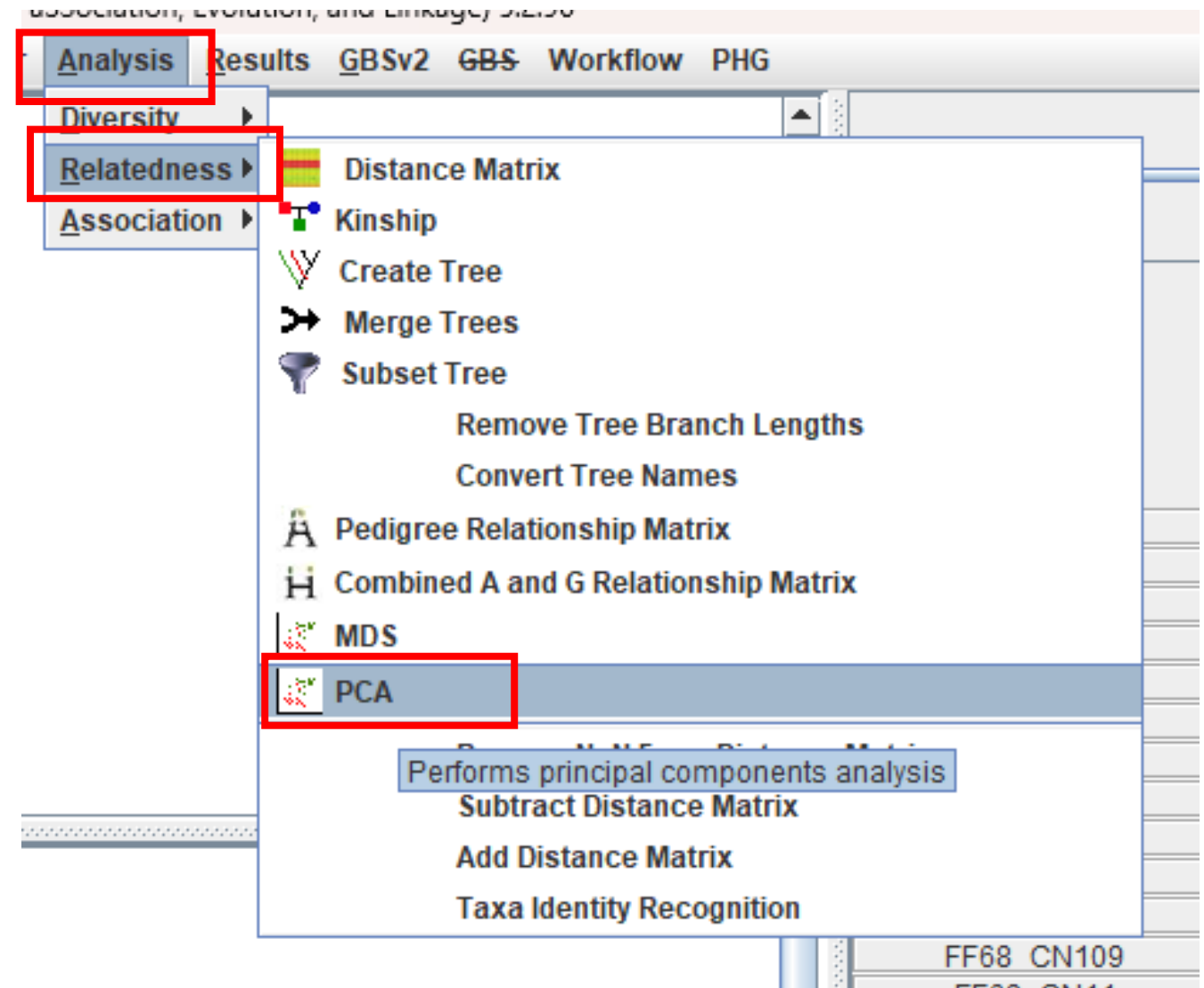
Table Title: trait
Number of columns: 2
Number of rows: 170
Matrix size (excludes row headers): 170

PCA analysis

1. Click sequence (final_data)

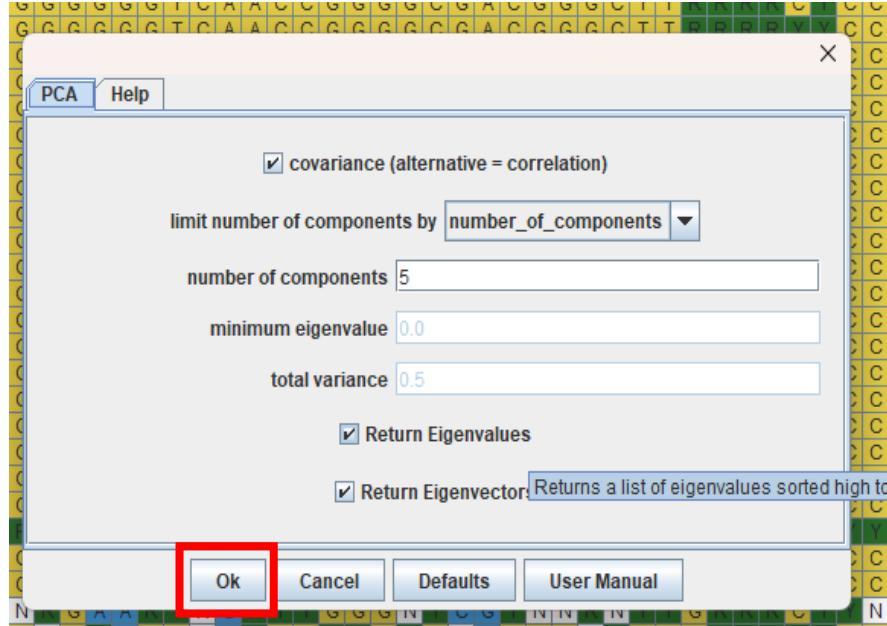


2. Analysis > Relatedness > PCA



PCA analysis

2. OK



3. Wait for result

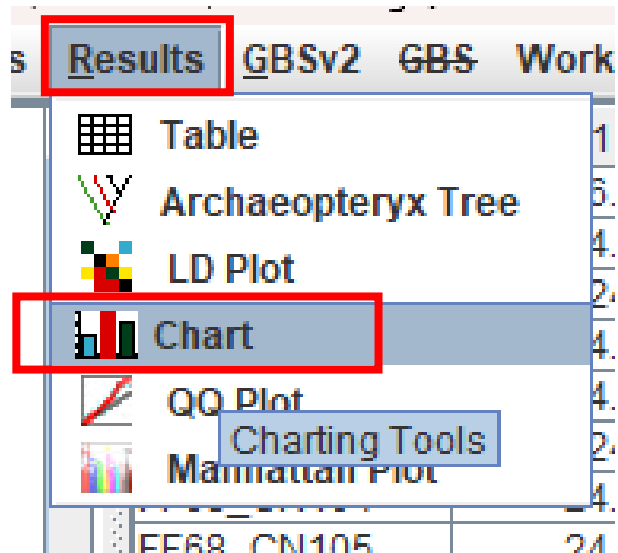
TASSEL (Trait Analysis by aSSociation, Evolution, and Linkage) 5.2.96

File Data Impute Filter Analysis Results GBSv2 GBS Workflow PHG

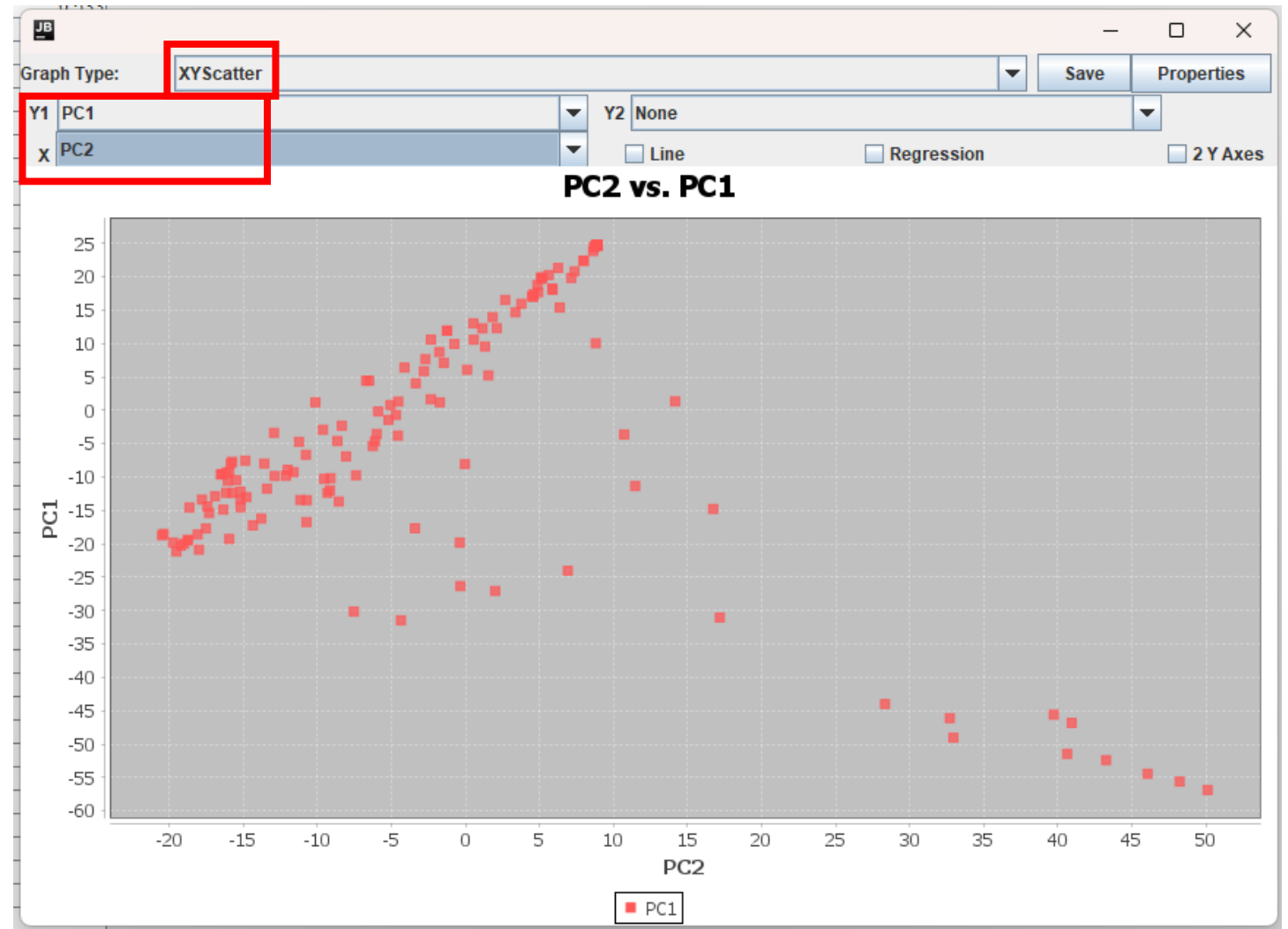
Taxa	PC1	PC2	PC3	PC4	PC5
FF68_CN1	16.529	2.684	1.989	-2.326	-8.392
FF68_CN10	24.748	8.919	1.587	0.432	0.509
FF68_CN100	24.73	8.903	1.609	0.43	0.465
FF68_CN101	24.737	8.879	1.606	0.45	0.516
FF68_CN102	24.735	8.887	1.584	0.442	0.489
FF68_CN103	24.73	8.907	1.59	0.44	0.511
FF68_CN104	24.745	8.895	1.603	0.407	0.467
FF68_CN105	24.437	8.678	1.588	0.478	0.465
FF68_CN106	24.755	8.908	1.613	0.449	0.517
FF68_CN107	24.76	8.907	1.616	0.447	0.533
FF68_CN108	24.768	8.904	1.618	0.448	0.535

PCA analysis

4. Results > Chart

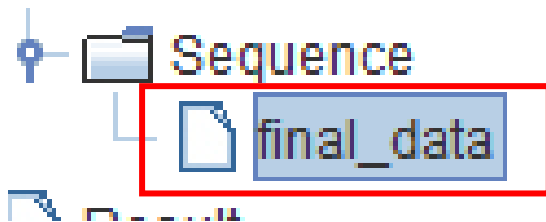


5. Plot PCA result by scatter plot

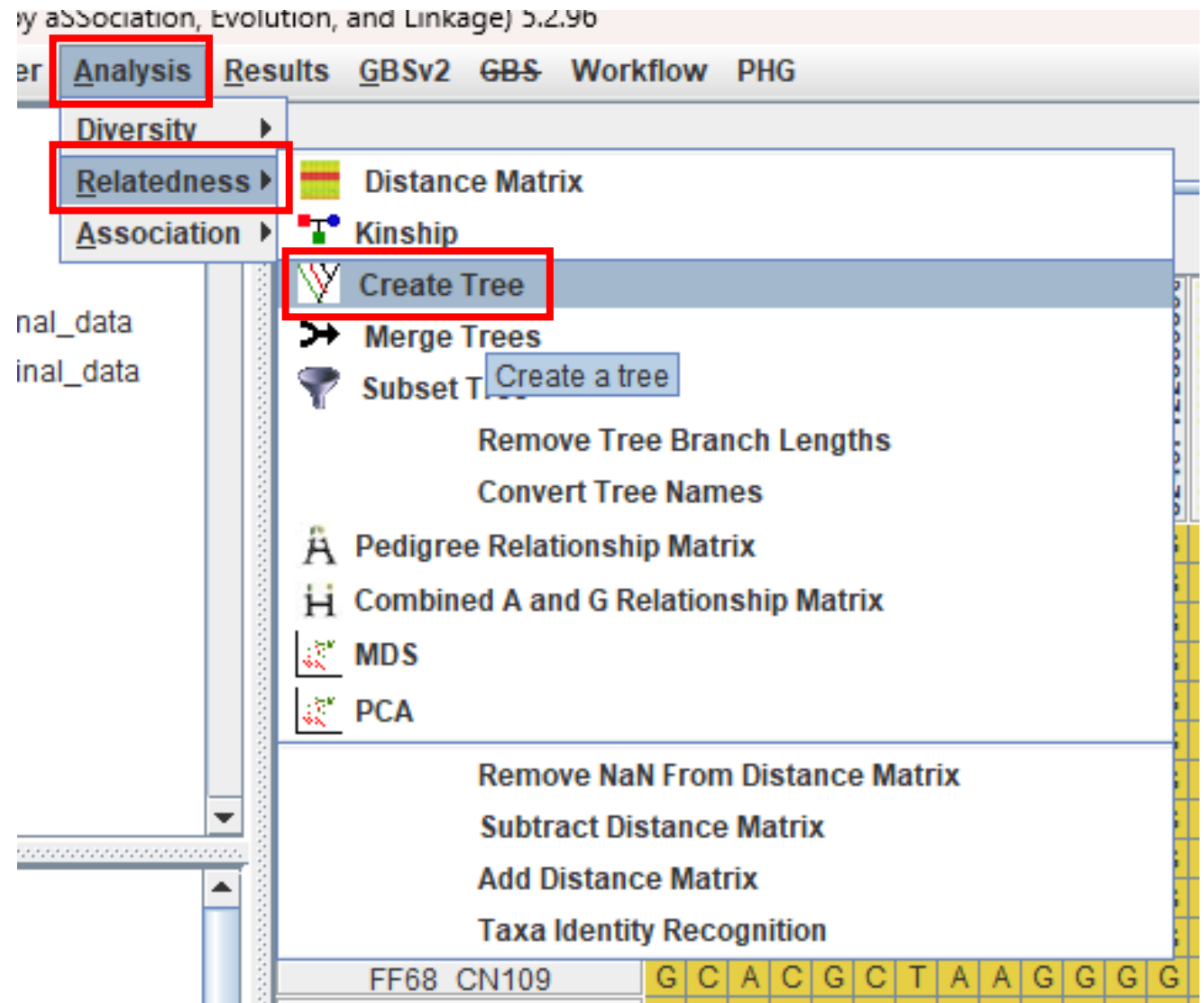


Phylogenetic analysis

1. Click sequence (final_data)

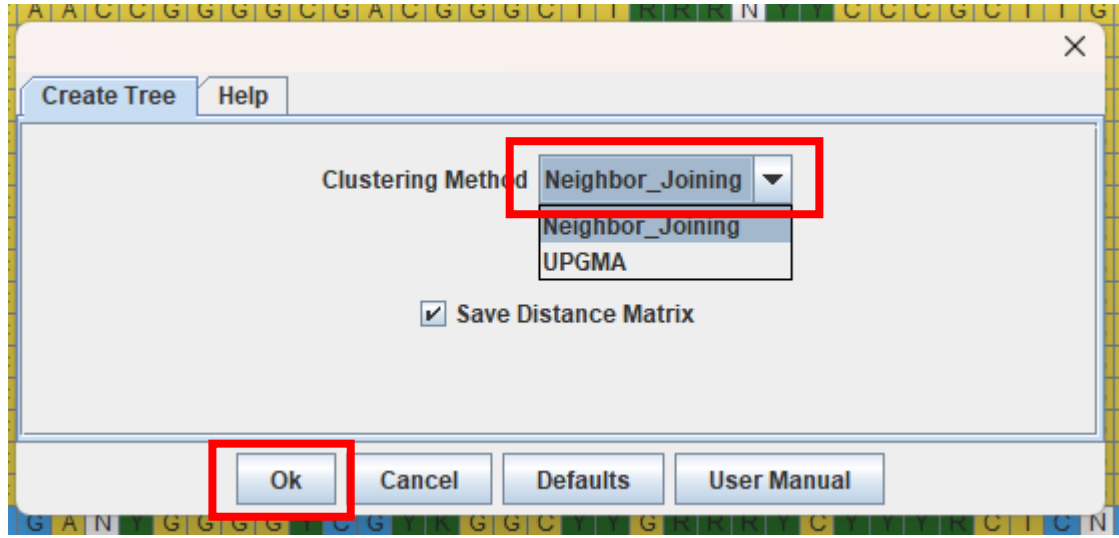


2. Analysis > Relatedness > Create Tree

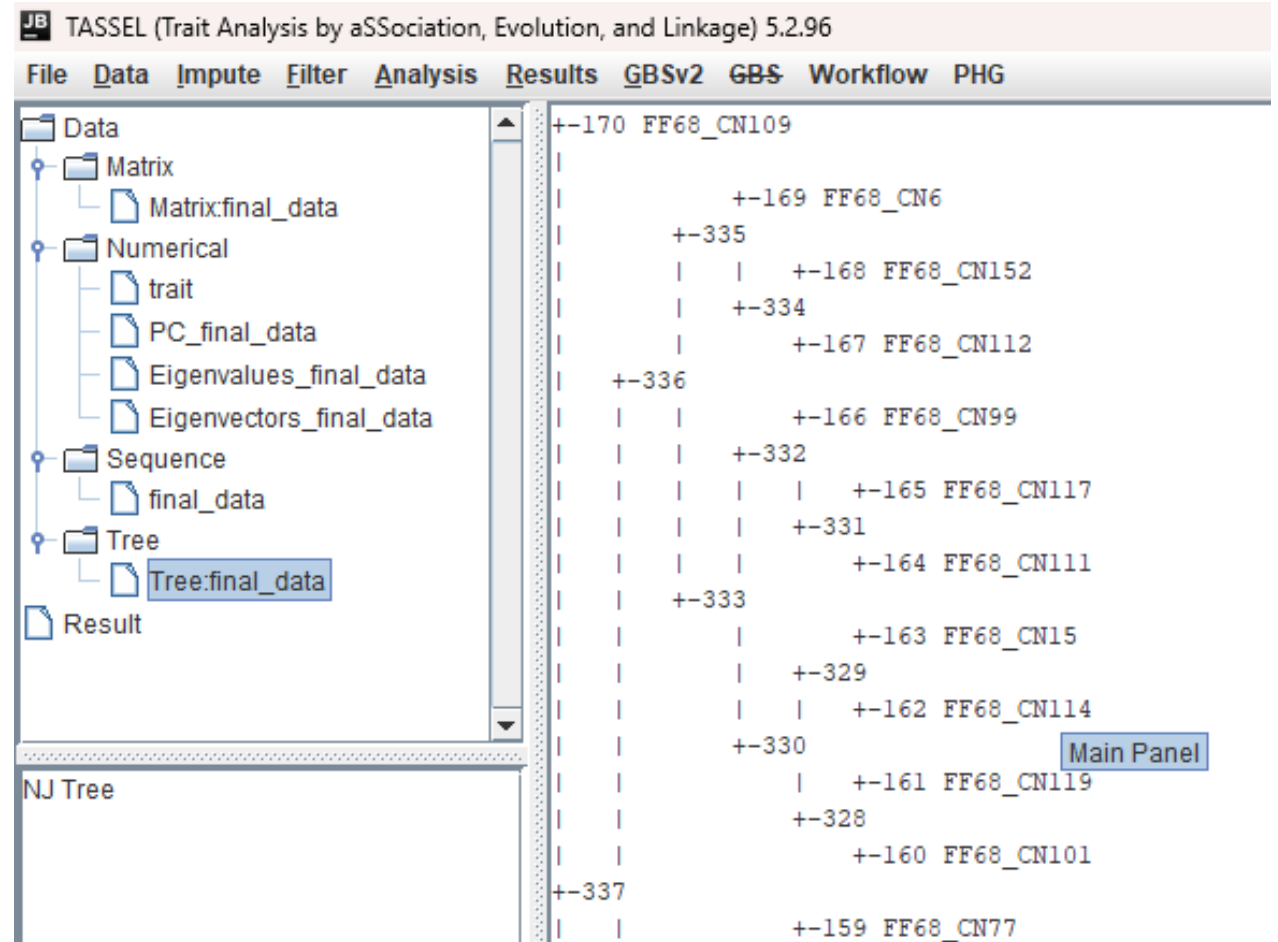


Phylogenetic analysis

3. Neighbor_Joining > Ok

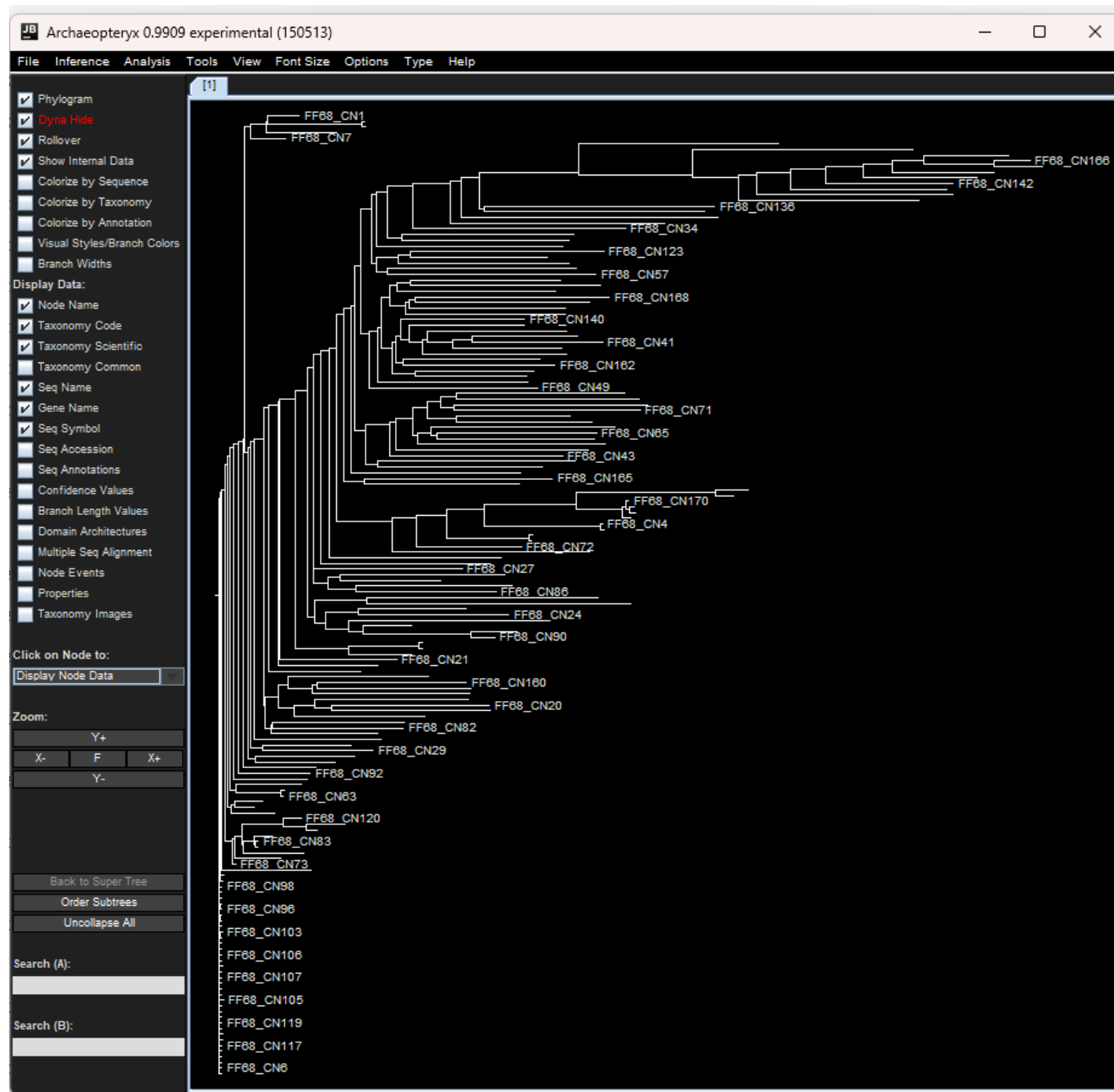
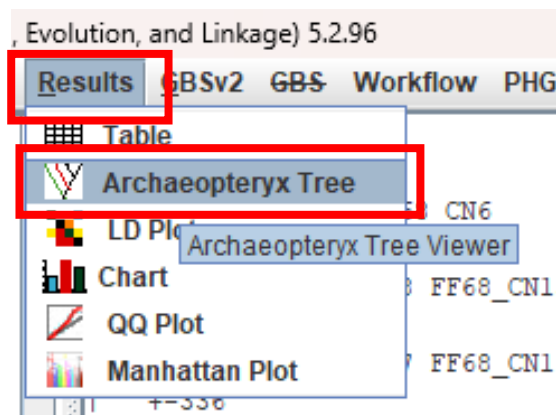


4. Wait for result



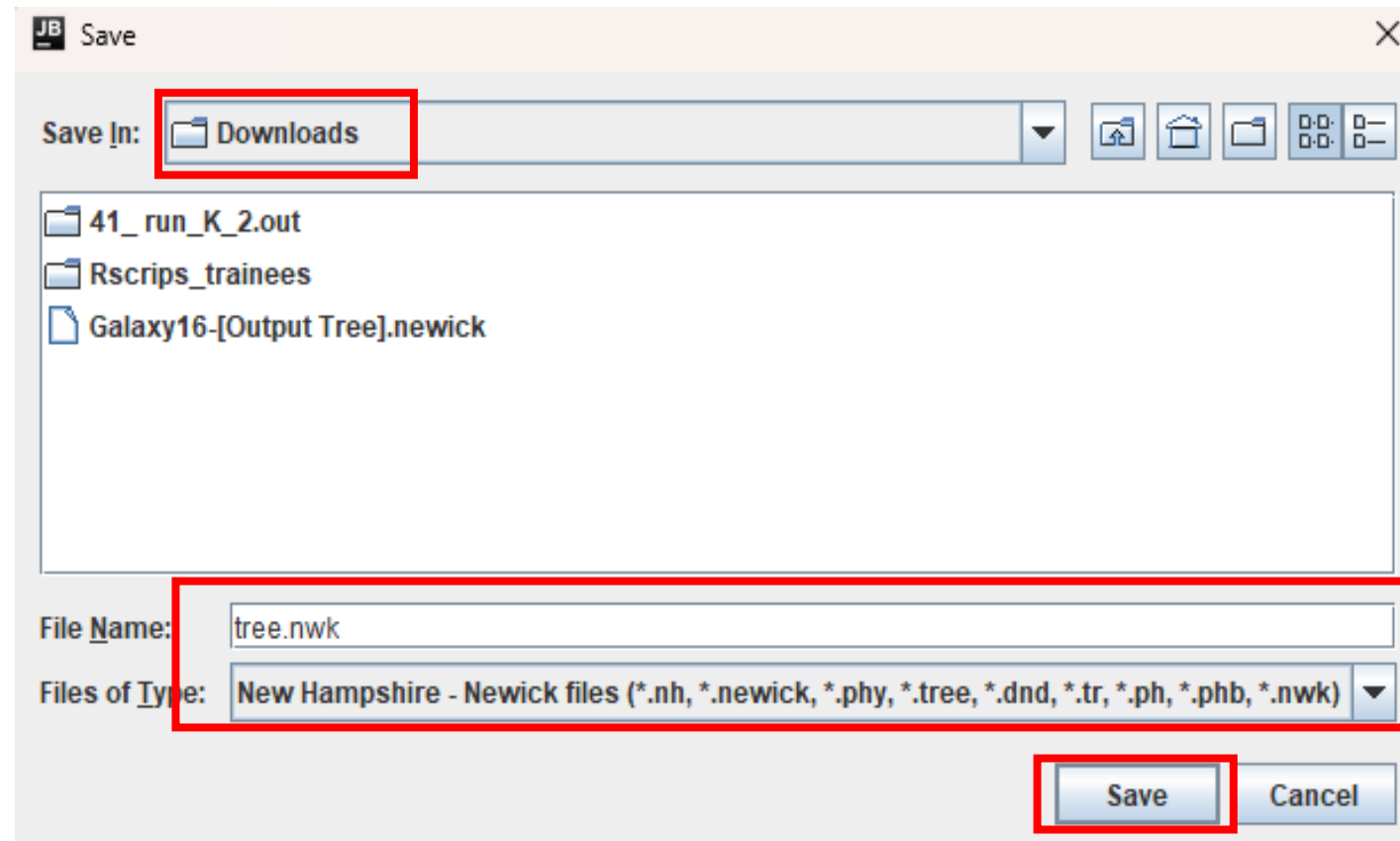
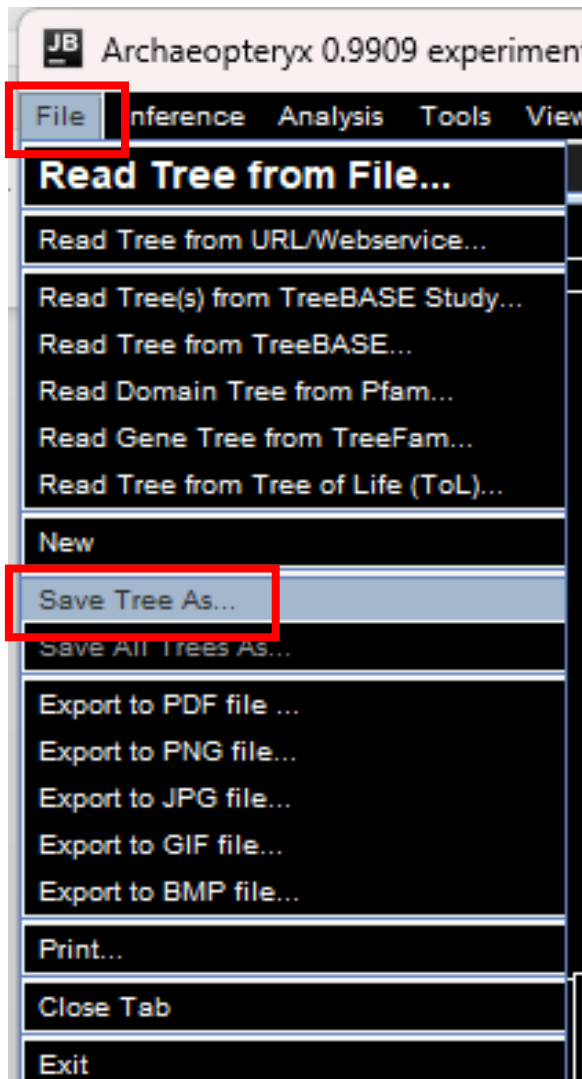
5. Plot Tree result

Result > Archaeopteryx Tree



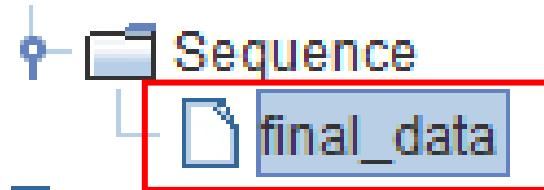
6. Export tree to .nwk file to plot other tools like Darwin or ITOL (<https://itol.embl.de/>)

File > Save Tree As > Location to save > set name to save > select Newick files > Save

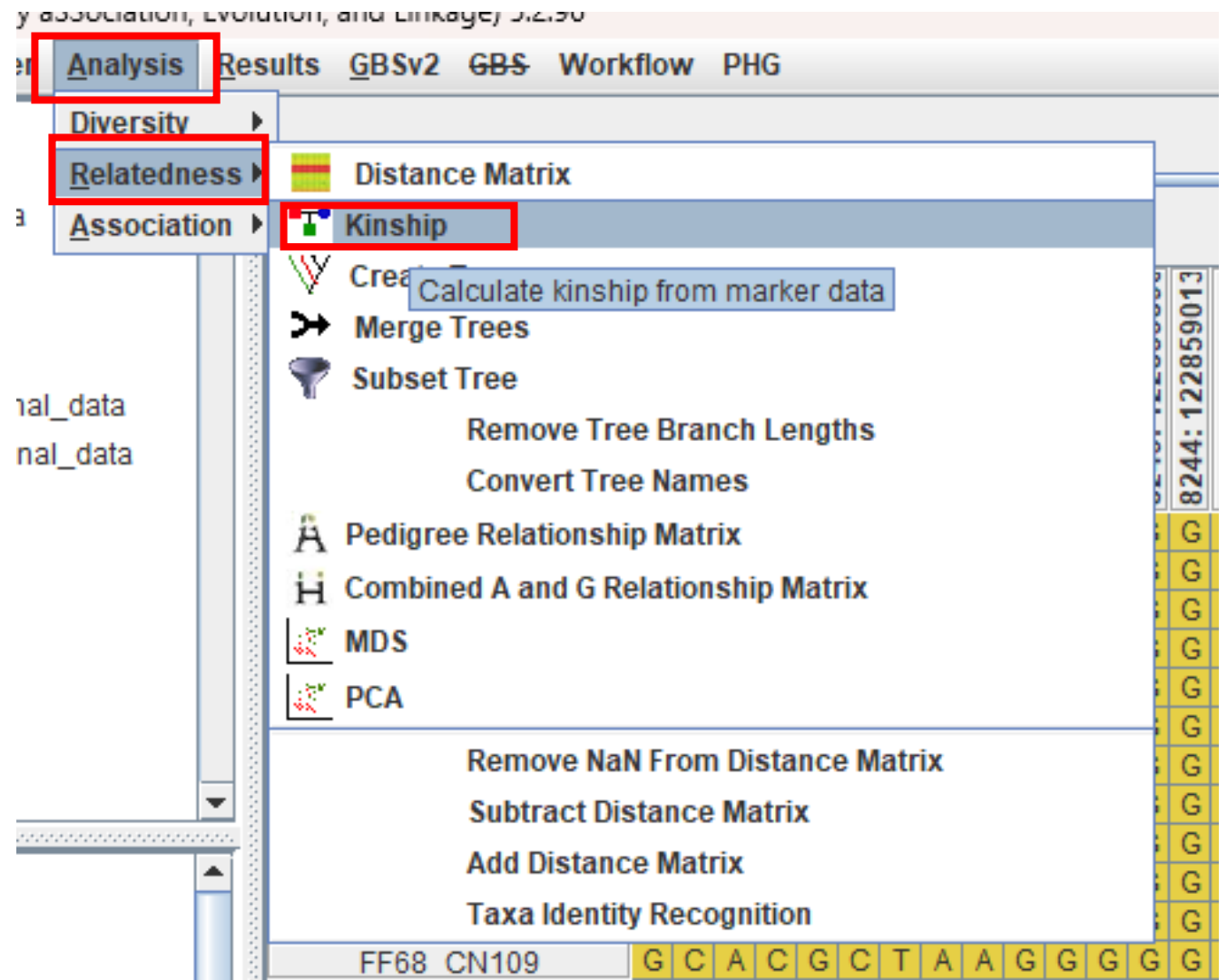


Kinship analysis (like populations structure)

1. Click sequence (final_data)



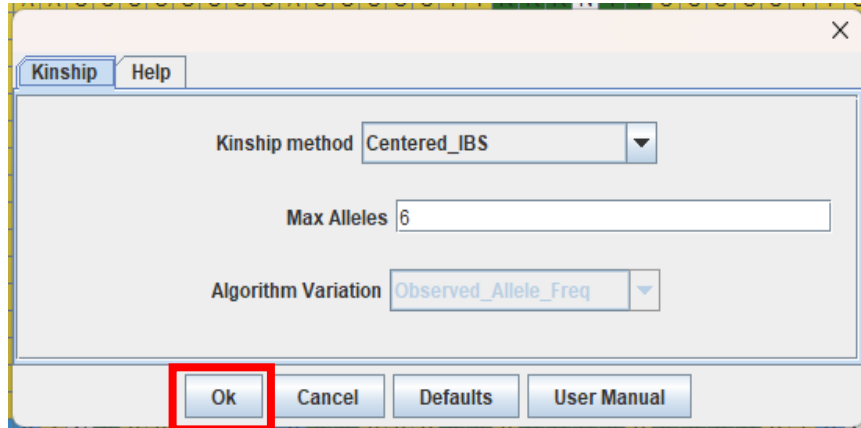
2. Analysis > Relatedness > Kinship



Kinship analysis (like populations structure)

4. Wait for result

3. OK



TASSEL (Trait Analysis by aSSociation, Evolution, and Linkage) 5.2.96

File Data Impute Filter Analysis Results GBSv2 GBS Workflow PHG

Taxa	FF68_CN1	FF68_CN10	FF68_CN1...	FF68_CN1...	FF68_CN1...	FF68_CN1...	FF68_CN1...	FF68_CN1...	FF68_CN1...
FF68_CN1	0.837	0.428	0.43	0.428	0.428	0.428	0.43	0.42	0.427
FF68_CN10	0.428	0.712	0.704	0.705	0.704	0.705	0.703	0.687	0.705
FF68_CN100	0.43	0.704	0.712	0.706	0.704	0.705	0.704	0.687	0.706
FF68_CN101	0.428	0.705	0.706	0.714	0.706	0.707	0.706	0.689	0.707
FF68_CN102	0.428	0.704	0.704	0.706	0.712	0.705	0.704	0.688	0.706
FF68_CN103	0.428	0.705	0.705	0.707	0.705	0.713	0.705	0.688	0.707
FF68_CN104	0.43	0.703	0.704	0.706	0.704	0.705	0.711	0.687	0.706
FF68_CN105	0.42	0.687	0.687	0.689	0.688	0.688	0.687	0.699	0.689
FF68_CN106	0.427	0.705	0.706	0.707	0.706	0.707	0.706	0.689	0.713
FF68_CN107	0.428	0.706	0.707	0.709	0.707	0.708	0.707	0.69	0.708
FF68_CN108	0.427	0.704	0.705	0.707	0.705	0.706	0.705	0.688	0.707
FF68_CN109	0.429	0.705	0.705	0.707	0.705	0.706	0.705	0.689	0.707
FF68_CN11	0.424	0.694	0.695	0.696	0.695	0.696	0.694	0.678	0.696
FF68_CN110	0.43	0.705	0.705	0.707	0.705	0.706	0.705	0.688	0.707
FF68_CN111	0.424	0.7	0.701	0.703	0.701	0.702	0.701	0.684	0.702
FF68_CN112	0.429	0.704	0.704	0.706	0.704	0.705	0.704	0.687	0.706
FF68_CN113	0.429	0.707	0.707	0.709	0.707	0.709	0.707	0.69	0.709
FF68_CN114	0.428	0.702	0.703	0.705	0.703	0.704	0.702	0.685	0.704
FF68_CN115	0.428	0.704	0.704	0.706	0.704	0.705	0.704	0.687	0.706
FF68_CN116	0.426	0.7	0.7	0.703	0.7	0.702	0.7	0.684	0.702
FF68_CN117	0.429	0.705	0.705	0.707	0.706	0.707	0.705	0.689	0.707
FF68_CN118	0.174	0.441	0.442	0.443	0.443	0.442	0.44	0.43	0.443
FF68_CN119	0.43	0.706	0.706	0.708	0.706	0.707	0.706	0.689	0.708
FF68_CN12	0.49	0.263	0.265	0.263	0.263	0.264	0.266	0.259	0.264
FF68_CN120	0.215	0.444	0.444	0.445	0.443	0.446	0.445	0.432	0.446
FF68_CN121	0.314	0.529	0.529	0.53	0.528	0.531	0.529	0.514	0.531
FF68_CN122	-0.559	-0.584	-0.583	-0.585	-0.584	-0.584	-0.586	-0.574	-0.584
FF68_CN123	-0.377	-0.417	-0.417	-0.418	-0.417	-0.416	-0.417	-0.406	-0.418
FF68_CN124	0.003	0.122	0.123	0.123	0.123	0.124	0.124	0.121	0.124
FF68_CN125	-0.38	-0.48	-0.484	-0.483	-0.484	-0.482	-0.483	-0.471	-0.482
FF68_CN126	-0.27	-0.364	-0.367	-0.365	-0.364	-0.365	-0.366	-0.355	-0.365

Table Title: Distance Matrix
Number of columns: 171
Number of rows: 170
Matrix size (excludes row headers): 28900

Annotations...

Centered_IBS.SumPc:
4036.8057937268727

Matrix_Type: Centered_IBS

Centered_IBS matrix created from
final_data

GWAS analysis

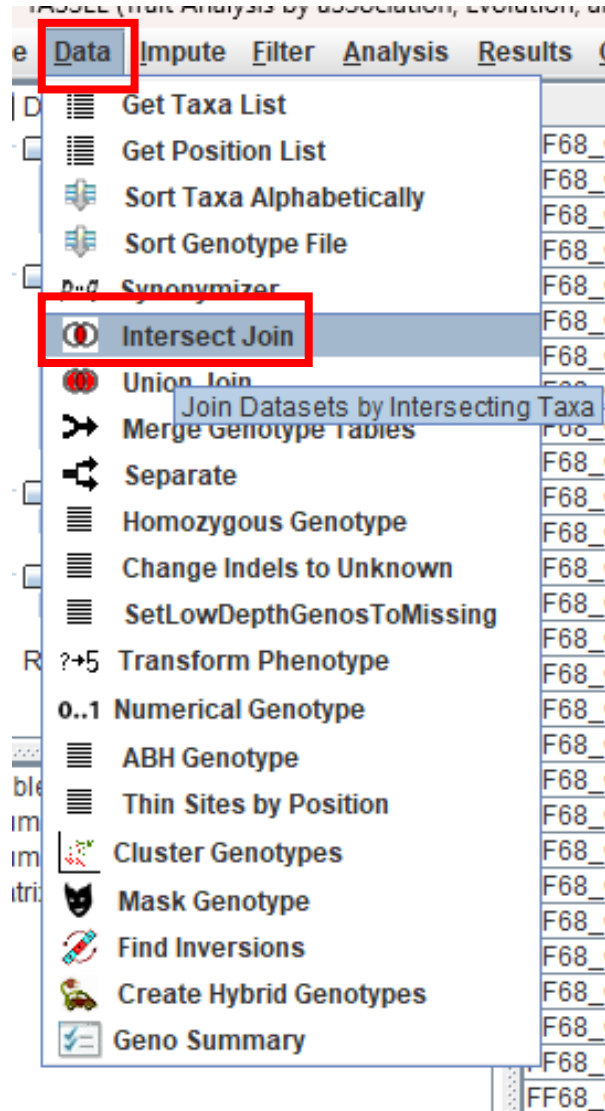
1. Merge PCA / trait / sequence data together (trait + PC_final_data + final_data)
 - Ctrl + click at trait
 - Hold Ctrl + click at PC_final_data
 - Hold Ctrl + final_data

The screenshot shows the TASSEL (Trait Analysis by Association, Evolution, and Linkage) 3.2.30 interface. The 'Data' panel on the left displays a hierarchical tree of data files. The 'Matrix' folder contains 'Matrix:final_data' and 'Centered_IBS_final_data'. The 'Numerical' folder contains 'trait', 'PC_final_data', 'Eigenvalues_final_data', and 'Eigenvectors_final_data'. The 'Sequence' folder contains 'final_data'. The 'Tree' folder contains 'Tree:final_data'. The 'Result' folder is empty. The 'trait' and 'PC_final_data' files are highlighted with a red box, and the 'final_data' file is also highlighted with a red box. The 'Table Title: trait' section at the bottom left shows the following information: Number of columns: 2, Number of rows: 170, Matrix size (excludes row headers): 170. The 'Results' panel on the right displays a table with two columns: 'Taxa' and 'Color'. The table contains 24 rows of data, with the first row being 'FF68_CN1' and the last row being 'FF68_CN24'. The 'Color' column contains values ranging from 1 to 5, with some rows having a diamond symbol in the 'Color' column.

Taxa	Color
FF68_CN1	1
FF68_CN2	1
FF68_CN3	3
FF68_CN4	5
FF68_CN5	5
FF68_CN6	1
FF68_CN7	1
FF68_CN8	1
FF68_CN9	1
FF68_CN10	1
FF68_CN11	1
FF68_CN12	1
FF68_CN13	2
FF68_CN14	5
FF68_CN15	1
FF68_CN16	1
FF68_CN17	1
FF68_CN18	1
FF68_CN19	◆
FF68_CN20	1
FF68_CN21	1
FF68_CN22	1
FF68_CN23	2
FF68_CN24	1

GWAS analysis

2. Data > Intersect Join



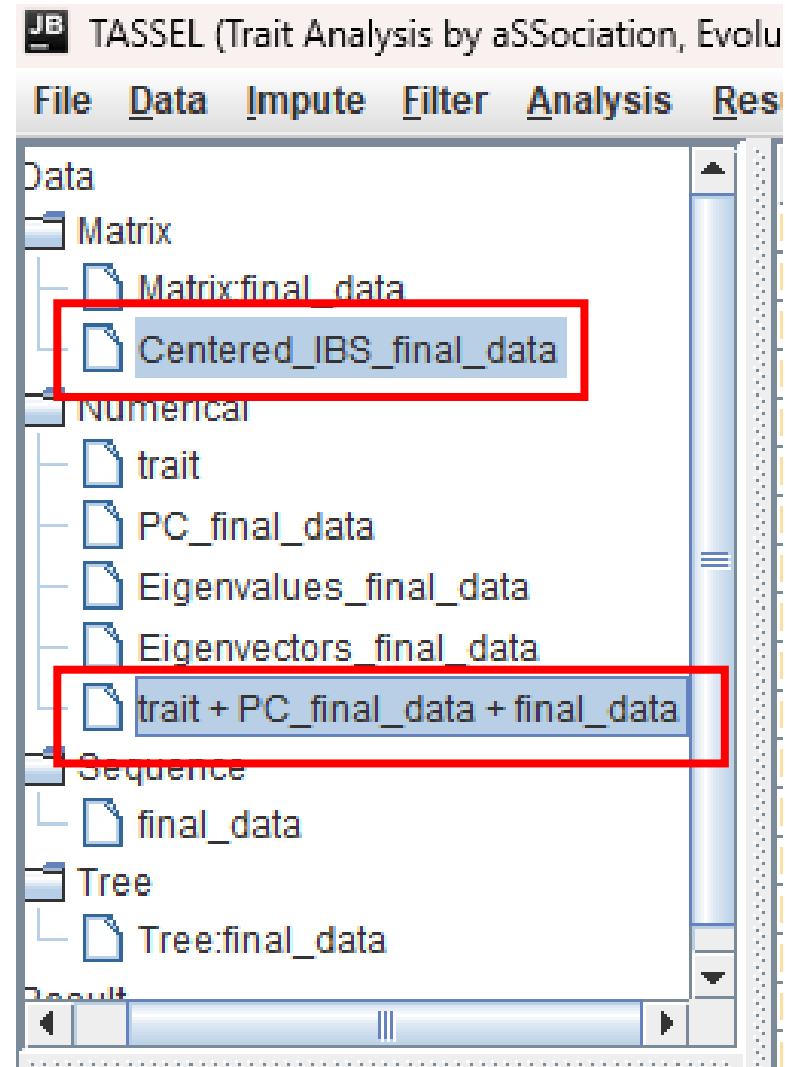
3. Got new merged table

The screenshot shows the 'Results' tab of the software interface. A table is displayed with the following columns: Taxa, Color, PC1, PC2, PC3, PC4, PC5, and Genotype. The table contains 10 rows of data, each representing a different taxon (FF68_CN1 to FF68_CN108). The 'Color' column has a value of 1 for all rows. The 'PC' columns contain numerical values. The 'Genotype' column contains the sequence 'CAAAGTG...' for all rows. A red box highlights the table header and the first row of data.

Taxa	Color	PC1	PC2	PC3	PC4	PC5	Genotype
FF68_CN1	1	16.529	2.684	1.989	-2.326	-8.392	CAAAGTG...
FF68_CN10	1	24.748	8.919	1.587	0.432	0.509	CAAAGTG...
FF68_CN100	1	24.73	8.903	1.609	0.43	0.465	CAAAGTG...
FF68_CN101	1	24.737	8.879	1.606	0.45	0.516	CAAAGTG...
FF68_CN102	1	24.735	8.887	1.584	0.442	0.489	CAAAGTG...
FF68_CN103	1	24.73	8.907	1.59	0.44	0.511	CAAAGTG...
FF68_CN104	1	24.745	8.895	1.603	0.407	0.467	CAAAGTG...
FF68_CN105	1	24.437	8.678	1.588	0.478	0.465	CAAAGTG...
FF68_CN106	1	24.755	8.908	1.613	0.449	0.517	CAAAGTG...
FF68_CN107	1	24.76	8.907	1.616	0.447	0.533	CAAAGTG...
FF68_CN108	1	24.762	8.884	1.618	0.406	0.489	CAAAGTG...

GWAS analysis

4. Selected tabel (trait + PC_final_data + final_data)
 - Hold Ctrl + click at Centered_IBS_final_data



GWAS analysis

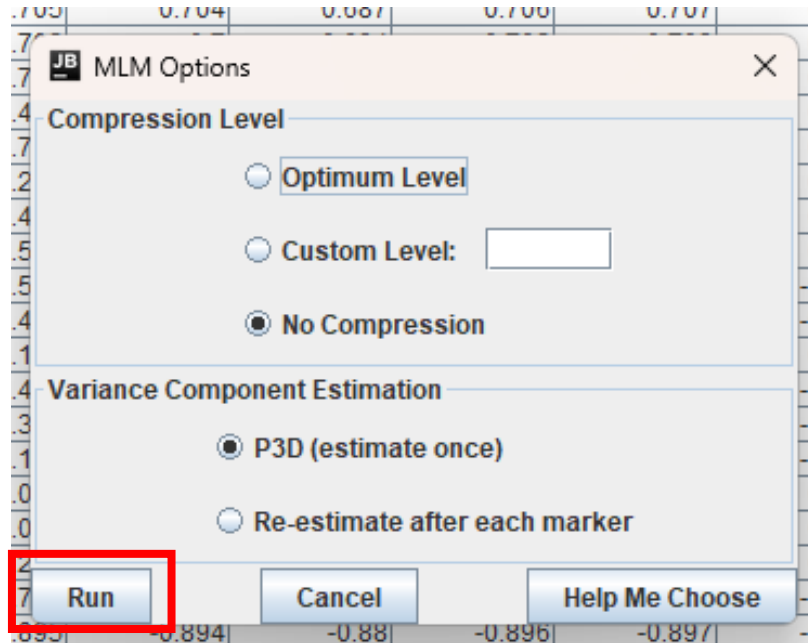
5. Analysis > Association > MLM

The screenshot shows the GBSv2 software interface. The 'Analysis' menu is open, and the 'Association' option is selected, which has opened a sub-menu. In this sub-menu, 'GLM' and 'MLM' are highlighted with red boxes. A tooltip for 'MLM' is visible, stating 'Association analysis using mixed model'. The background shows a file tree on the left and a data table on the right.

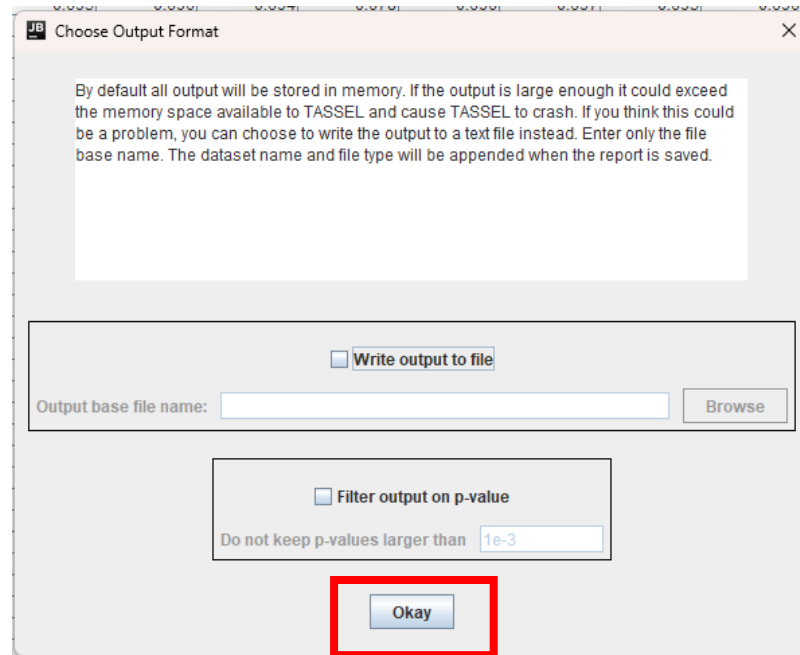
Taxa	FF68_CN1	FF68_CN10	FF68_CN11
FF68_CN1	0.837	0.428	0.428
FF68_CN10			0.705
FF68_CN11			0.705
FF68_CN12			0.705
FF68_CN13			0.705
FF68_CN14			0.705

GWAS analysis

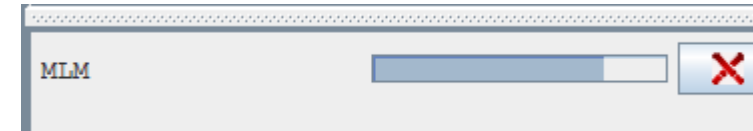
5. Run



6. Okay



7. Waiting...



GWAS analysis

8. Got result table of GWAS analysis

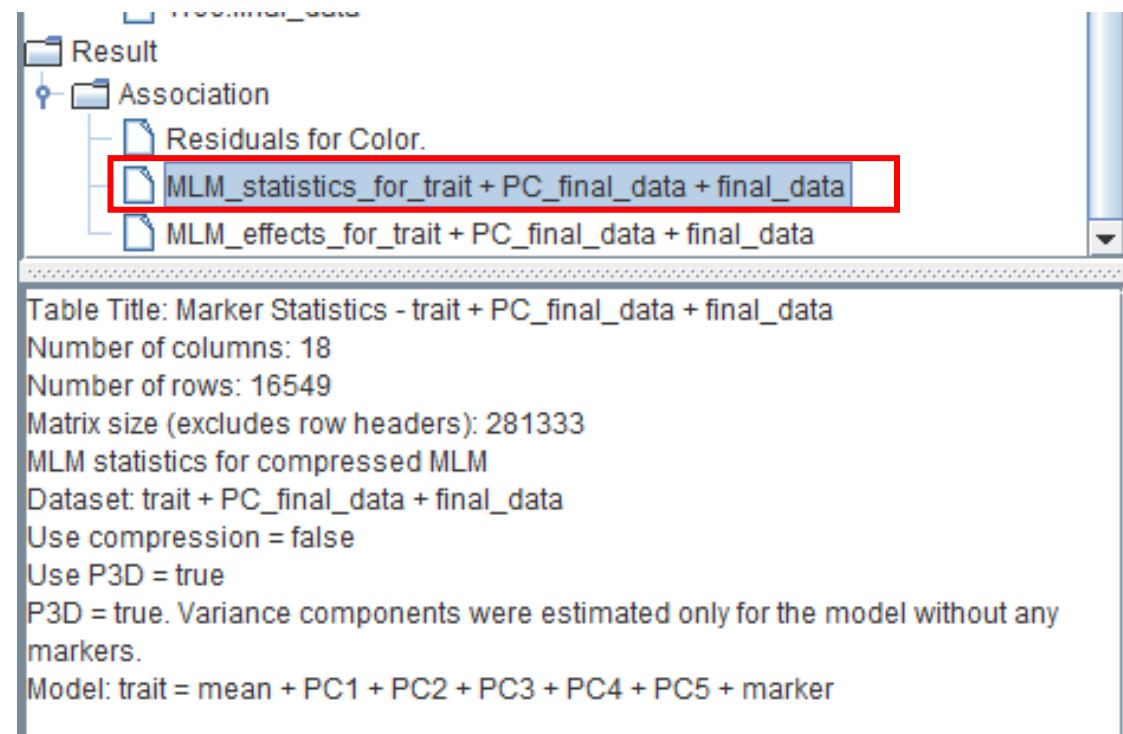


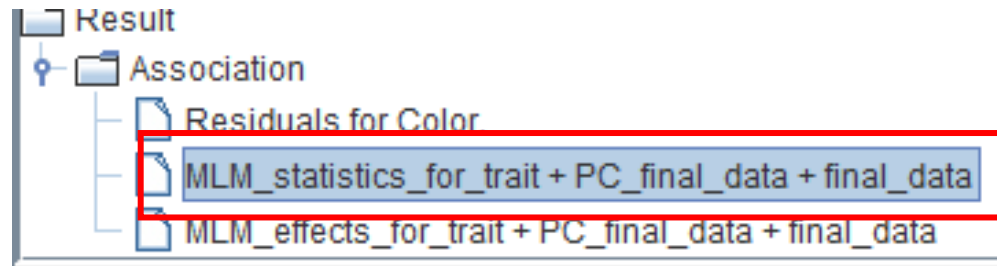
Table Title: Marker Statistics - trait + PC_final_data + final_data
 Number of columns: 18
 Number of rows: 16549
 Matrix size (excludes row headers): 281333
 MLM statistics for compressed MLM
 Dataset: trait + PC_final_data + final_data
 Use compression = false
 Use P3D = true
 P3D = true. Variance components were estimated only for the model without any markers.
 Model: trait = mean + PC1 + PC2 + PC3 + PC4 + PC5 + marker

Trait	Marker	Chr	Pos	df	F	p	add_effect	add_F	add_p	dom_effect	dom_F	dom_p	errordf	MarkerR2	Genetic Var	Residual Var	-2LnLikelih...
Color	None			0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	162	NaN	0.69173	6.9173E-6	138.08855
Color	S10_1016...	10	101648167	2	0.06234	0.93959	0.21014	0.12461	0.72458	0.01973	0.00381	0.95089	160.0	7.4332E-4	0.69173	6.9173E-6	138.08855
Color	S10_1016...	10	101663454	2	0.16546	0.84765	0.18173	0.10939	0.74128	0.15323	0.24971	0.61798	164.0	0.00195	0.69173	6.9173E-6	138.08855
Color	S10_1016...	10	101688699	2	0.6273	0.53538	-5.6295E-2	0.03124	0.85993	0.43794	1.15007	0.2852	163.0	0.00749	0.69173	6.9173E-6	138.08855
Color	S10_1016...	10	101698798	2	1.82157	0.16531	0.02979	0.00849	0.9267	0.34204	3.53255	0.0621	159.0	0.02195	0.69173	6.9173E-6	138.08855
Color	S10_1017...	10	101703868	2	1.92525	0.14931	0.14792	0.17607	0.67535	0.38616	3.02877	0.08378	163.0	0.02305	0.69173	6.9173E-6	138.08855
Color	S10_1017...	10	101708910	2	0.08249	0.92086	-1.1629E-1	0.04724	0.82822	0.09355	0.12173	0.72763	167.0	9.6449E-4	0.69173	6.9173E-6	138.08855
Color	S10_1017...	10	101749796	2	0.08352	0.91991	-7.7035E-2	0.07915	0.77882	-4.527E-2	0.02685	0.87005	166.0	9.8177E-4	0.69173	6.9173E-6	138.08855
Color	S10_1017...	10	101765041	2	2.28432	0.10524	0.20115	0.15236	0.69682	0.35724	3.16335	0.07726	164.0	0.02774	0.69173	6.9173E-6	138.08855
Color	S10_1017...	10	101770144	2	0.12621	0.88152	0.18468	0.23771	0.62654	-1.9942E-2	0.00499	0.94376	166.0	0.00149	0.69173	6.9173E-6	138.08855
Color	S10_1017...	10	101775146	2	0.09379	0.91052	0.04545	0.01029	0.91932	0.11952	0.18025	0.67173	166.0	0.0011	0.69173	6.9173E-6	138.08855
Color	S10_1018...	10	101805528	2	0.57936	0.56145	0.25188	0.47845	0.49015	-2.1974E-1	0.66179	0.41716	165.0	0.00683	0.69173	6.9173E-6	138.08855
Color	S10_1018...	10	101810838	2	0.08749	0.91628	-4.1722E-2	0.01046	0.91868	0.10843	0.14953	0.6995	166.0	0.00103	0.69173	6.9173E-6	138.08855
Color	S10_1018...	10	101815892	2	2.65276	0.07361	0.43296	0.42905	0.51341	0.32515	3.58631	0.0601	165.0	0.0313	0.69173	6.9173E-6	138.08855
Color	S10_1018...	10	101820982	2	0.03887	0.96188	0.02052	0.0019	0.96528	0.07122	0.07337	0.78684	166.0	4.572E-4	0.69173	6.9173E-6	138.08855
Color	S10_1018...	10	101826180	2	0.14541	0.86478	0.16096	0.1056	0.74565	0.12928	0.20704	0.64973	164.0	0.00182	0.69173	6.9173E-6	138.08855
Color	S10_1018...	10	101831207	2	2.10836	0.12484	0.17664	0.25049	0.61743	0.39573	4.21359	0.04175	166.0	0.02479	0.69173	6.9173E-6	138.08855

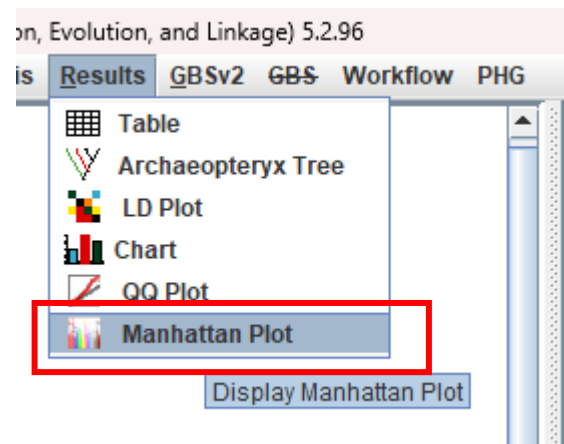
GWAS analysis

Manhattan plot for GWAS visualize result

1. Select GWAS result table (MLM_statistics_for_trait + PC_final_data + final_data)

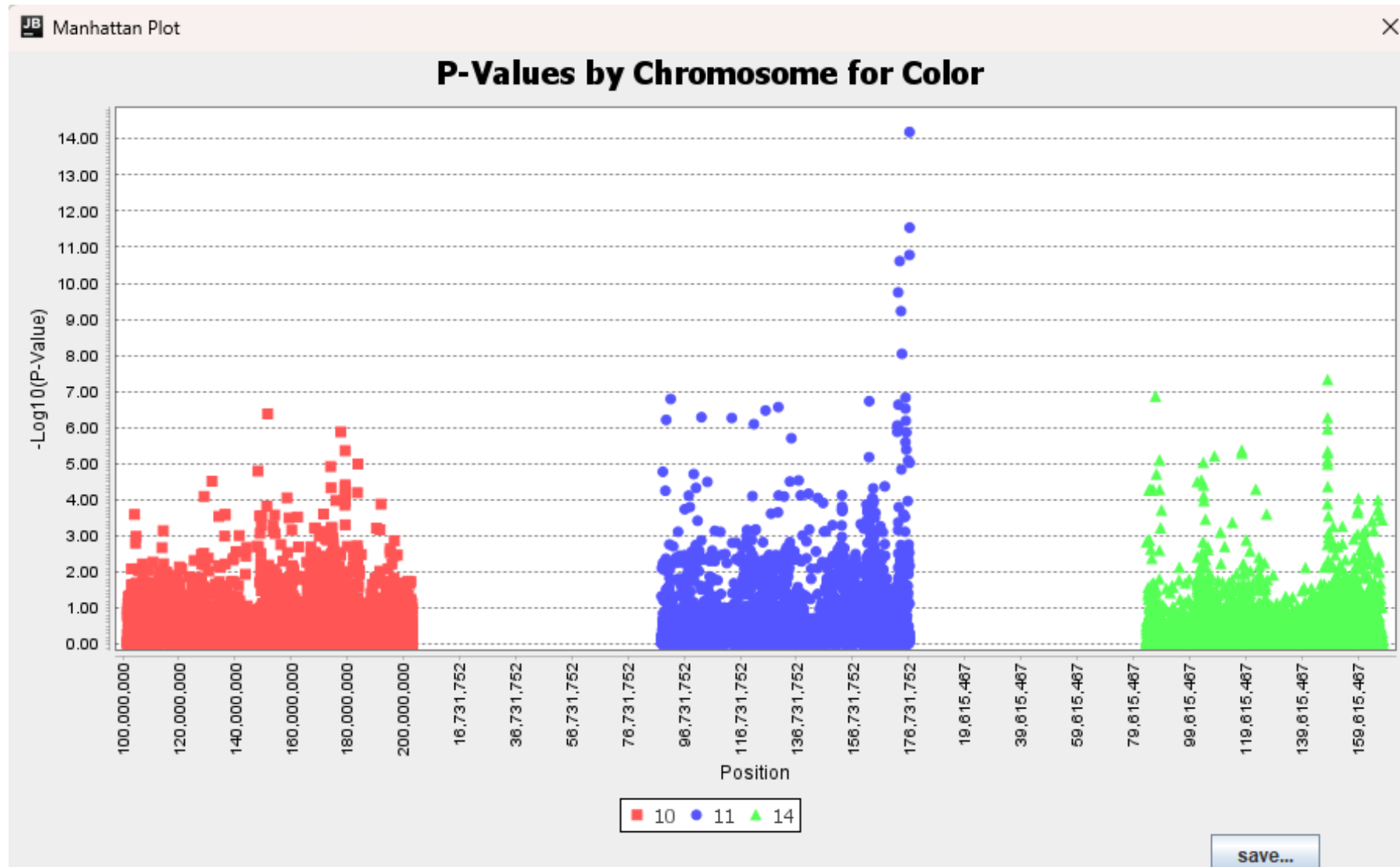


2. Results > Manhattan plot



GWAS analysis

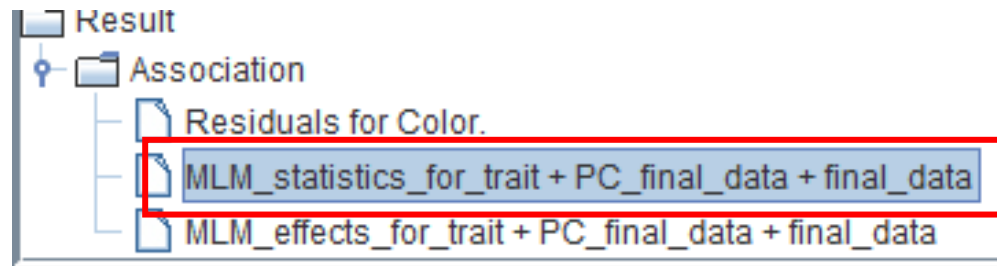
Manhattan plot for GWAS visualized result



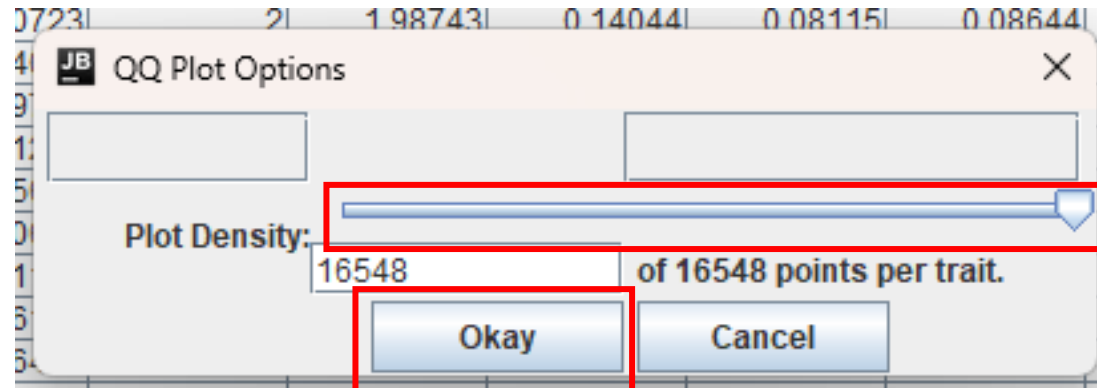
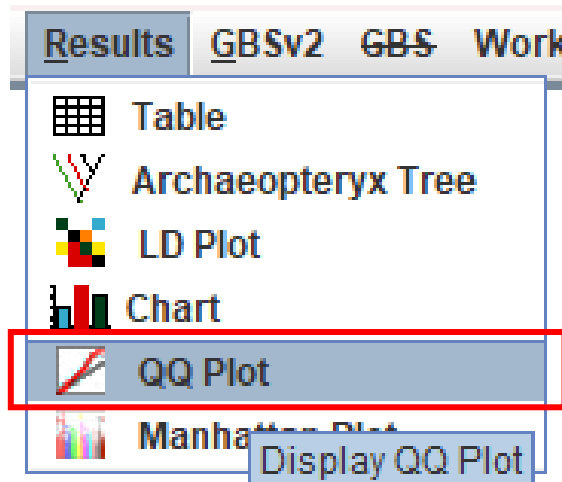
GWAS analysis

QQ plot

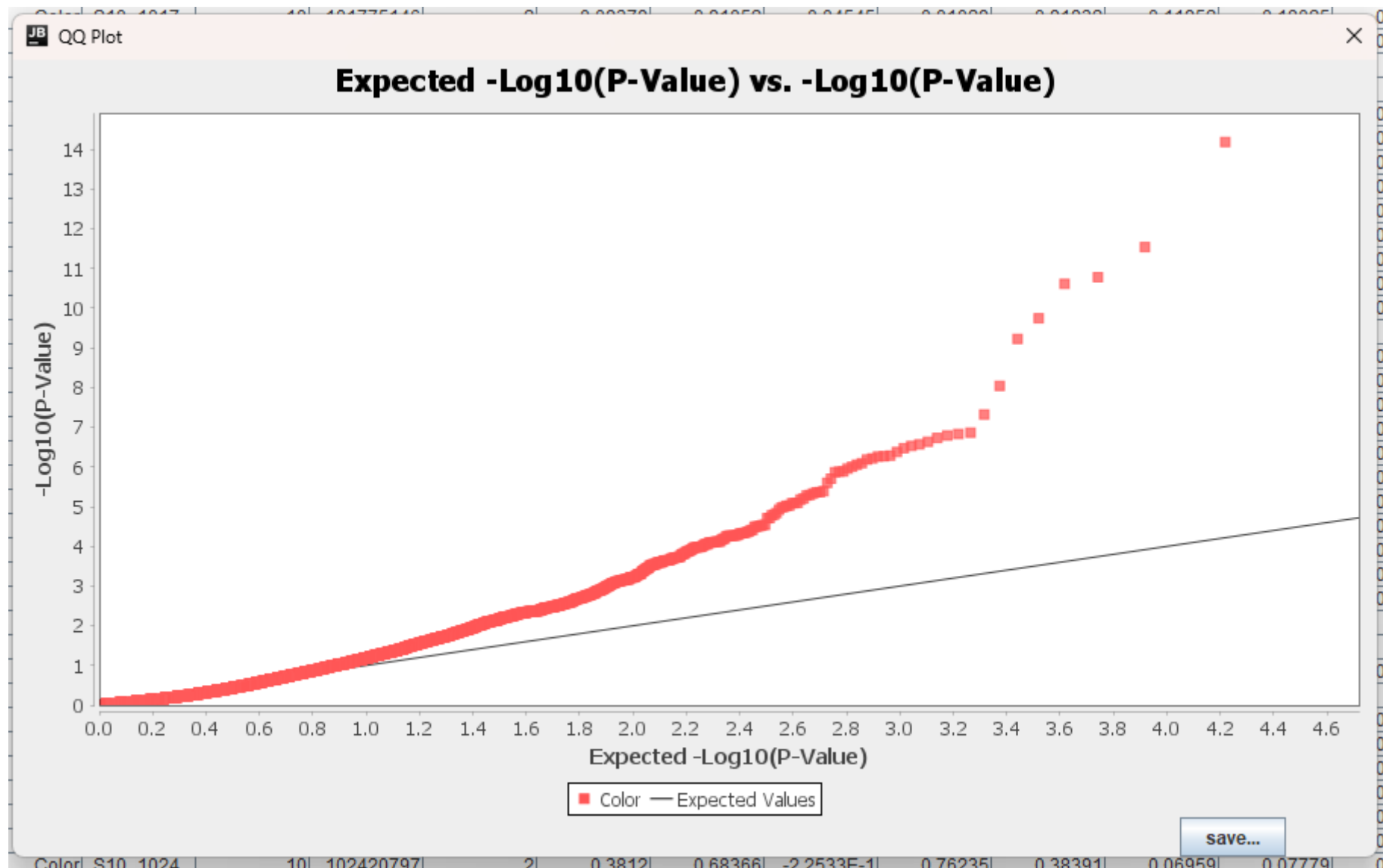
1. Select GWAS result table (MLM_statistics_for_trait + PC_final_data + final_data)



2. Results > QQ plot > select density SNP to plot > Okay



QQ plot of GWAS result

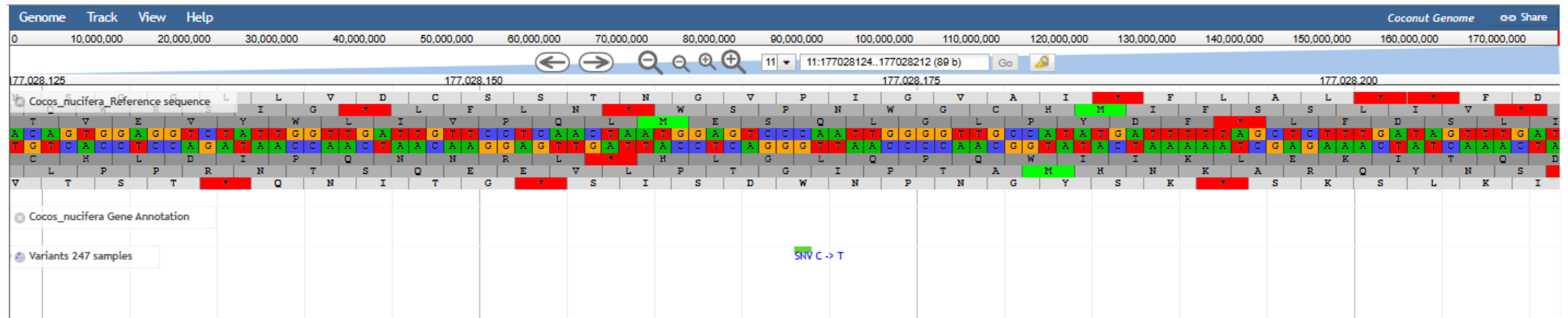


Genome browser for Coconut

Trait	Marker	Chr	Pos	df	F	p	add_effect	add_F	add_p	dom_effect	dom_F	dom_p	errordf	MarkerR2	Genetic Var	Residual Var	-2LnLikelih...
Color	S11_177028168	11	177028168	2	40.40761	6.4695E-15	-1.4676E0	78.03406	1.7787E-15	-9.3942E-1	14.73706	1.7805E-4	167.0	0.47245	0.69173	6.9173E-6	138.08855
Color	S11_177009216	11	177009216	2	29.39669	1.65E-11	1.13367	55.08657	7.8123E-12	-8.7006E-1	14.90302	1.6736E-4	159.0	0.35145	0.69173	6.9173E-6	138.08855
Color	S11_173532779	11	173532779	2	28.67082	2.4336E-11	1.1624	46.743	1.6958E-10	-1.3792E0	38.90084	3.9555E-9	165.0	0.35499	0.69173	6.9173E-6	138.08855
Color	S11_172977671	11	172977671	2	25.92995	1.7929E-10	1.22879	50.28471	4.1162E-11	-1.0206E0	22.93329	3.8133E-6	167.0	0.30317	0.69173	6.9173E-6	138.08855
Color	S11_173084044	11	173084044	2	24.42032	5.9328E-10	-1.1756E0	46.87384	1.6376E-10	-0.409E-1	10.80320	1.550E-6	164.0	0.28804	0.69173	6.9173E-6	138.08855

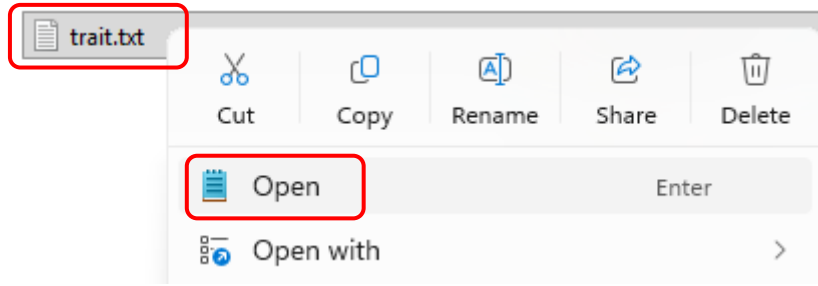
GWAS highly significant (lowest p-value) at Chr11:177028168

Go to <https://sinin.kps.ku.ac.th/ricedbtools/jbrowse/> for mining gene related with trait
(Chr11:177028168.. 177028168)



Edit phenotype header file

1. Open file with text editor



2. Change “Taxa” to <Trait>



Taxa	Color
FF68_CN1	1
FF68_CN2	1
FF68_CN3	3

<Trait>	Color
FF68_CN1	1
FF68_CN2	1
FF68_CN3	3

3. Save and Close

